CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

Introduction to Finite Elements in Engineering

Thoroughly updated with improved pedagogy, the fifth edition of this classic textbook continues to provide students with a clear and comprehensive introduction to the fundamentals of the finite element method. New features include coverage of core topics – including mechanics and heat conduction, energy and Galerkin approaches, convergence and adaptivity, time-dependent problems, and computer implementation – in the context of simple 1D problems, before advancing to 2D and 3D problems; expanded coverage of reduction of bandwidth, profile and fill-in for sparse solutions, time-dependent problems, plate bending, and nonlinearity; over 30 additional solved problems; and downloadable MATLAB, Python, C, JavaScript, Fortran, and Excel VBA code, providing students with hands-on experience. Accompanied by online solutions for instructors, this is the definitive text for senior undergraduate and graduate students studying a first course in the finite element method, and for professional engineers keen to shore up their understanding of finite element fundamentals.

Tirupathi R. Chandrupatla has been Professor and Founding Chair of Mechanical Engineering at Rowan University, New Jersey since 1995. He received his B.E. from the National Institute of Technology, Warangal, his M.Tech. from the Indian Institute of Technology (IIT), and his Ph.D. from the University of Texas, Austin. He started his career as a design engineer with Hindustan Machine Tools (HMT), Bangalore and then taught at IIT. Professor Chandrupatla also taught at the University of Kentucky and Kettering University, before joining Rowan. In 2005, he received the Lindback Distinguished Teaching Award at Rowan University. He is also the author of *Quality and Reliability in Engineering* (Cambridge, 2009) and *Introduction to Finite Elements in Engineering* (2012). Professor Chandrupatla has broad research interests, which include design, optimization, manufacturing engineering, finite element analysis, and quality and reliability. He has published widely in these areas and serves as a consultant to industry. Professor Chandrupatla is a registered Professional Engineer and also a Certified Manufacturing Engineer.

Ashok D. Belegundu has been Professor of Mechanical Engineering at Pennsylvania State University since 1986. Prior to this, he taught at Kettering University, Michigan. He received his B.Tech. from IIT Madras and his Ph.D. from the University of Iowa. He has been a principal investigator on research projects involving optimization for several agencies, including the National Science Foundation, Army Research Office, NASA, SERC (UK), MacNeal-Schwendler Corporation, Gentex Corporation, and Ingersoll-Rand. He has organized two international conferences on optimization in industry and has authored or edited four books and written a chapter in a book. He has advised more than 50 graduate students. He has given short courses on finite elements and optimization to the Forging Industry Association, Hazleton Pumps, and Infosys (India). He has served as an associate editor for *AIAA Journal* and for *Mechanics Based Design of Structures and Machines*.

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

Introduction to Finite Elements in Engineering

Fifth Edition

Tirupathi R. Chandrupatla
Rowan University, New Jersey

Ashok D. Belegundu
Pennsylvania State University

CAMBRIDGE
UNIVERSITY PRESS

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE
UNIVERSITY PRESS

Shaftesbury Road, Cambridge CB2 8EA, United Kingdom

One Liberty Plaza, 20th Floor, New York, NY 10006, USA

477 Williamstown Road, Port Melbourne, VIC 3207, Australia

314–321, 3rd Floor, Plot 3, Splendor Forum, Jasola District Centre, New Delhi – 110025, India

103 Penang Road, #05–06/07, Visioncrest Commercial, Singapore 238467

Cambridge University Press is part of Cambridge University Press & Assessment, a department of the University of Cambridge.

We share the University's mission to contribute to society through the pursuit of education, learning and research at the highest international levels of excellence.

www.cambridge.org
Information on this title: www.cambridge.org/9781108841412

DOI: 10.1017/9781108882293

© Ashok D. Belegundu and Tirupathi R. Chandrupatla 2022

This publication is in copyright. Subject to statutory exception and to the provisions of relevant collective licensing agreements, no reproduction of any part may take place without the written permission of Cambridge University Press & Assessment.

First published by Prentice-Hall 1990
Second edition 1996
Third edition 2002
Fourth edition 2012
Fifth edition by Cambridge University Press & Assessment 2022

A catalogue record for this publication is available from the British Library

Library of Congress Cataloging-in-Publication data
Names: Chandrupatla, Tirupathi R., 1944– author. | Belegundu, Ashok D., 1956– author.
Title: Introduction to finite elements in engineering / Tirupathi R. Chandrupatla, Rowan University, New Jersey, Ashok D. Belegundu, Pennsylvania State University.
Description: 5th edition. | Cambridge, United Kingdom ; New York, NY, USA : Cambridge University Press, 2022. | Includes bibliographical references.
Identifiers: LCCN 2021024618 (print) | LCCN 2021024619 (ebook) | ISBN 9781108841412 (hardback) | ISBN 9781108882293 (epub)
Subjects: LCSH: Finite element method. | Engineering mathematics.
Classification: LCC TA347.F5 C463 2022 (print) | LCC TA347.F5 (ebook) | DDC 620.001/51–dc23
LC record available at https://lccn.loc.gov/2021024618
LC ebook record available at https://lccn.loc.gov/2021024619

ISBN 978-1-108-84141-2 Hardback

Cambridge University Press & Assessment has no responsibility for the persistence or accuracy of URLs for external or third-party internet websites referred to in this publication and does not guarantee that any content on such websites is, or will remain, accurate or appropriate.

For
Suhasini and Uma

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

Contents

Detailed Contents *page* ix
Preface xix

1 Energy and Galerkin Approaches 1

2 Gaussian Elimination 35

3 One-Dimensional Elasticity 71

4 One-Dimensional Heat Conduction 118

5 Trusses 152

6 Two- and Three-Dimensional Elasticity 183

7 Axisymmetric Problems in Elasticity 259

8 Two- and Three-Dimensional Heat Conduction and Other Scalar Field Problems 293

9 Elements in Bending 340

10 Structural Vibration 384

11 Miscellaneous Topics 444

12 Preprocessing and Postprocessing 468

Appendix List of Symbols and Computer Code Variables 489
Bibliography 490
Index 493

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

Detailed Contents

Preface *page* xix

1 Energy and Galerkin Approaches 1

1.1 Introduction 1
1.2 Outline of Presentation 1
1.3 Overview of Types of Problems and Solution Approaches 2
1.3.1 Scalar Field Problems 3
1.3.2 Structural Mechanics 3
1.3.3 Solution Approaches 5
1.3.4 Decisions Involved in Defining the Problem 5
1.3.5 Consistent Units 6
1.4 Computer Programs 6
1.5 Energy Approach 7
1.5.1 Energy in a Spring System 7
1.5.2 Energy in a Rod Undergoing Axial Deformation 8
1.5.3 Energy in a Beam Undergoing Bending 9
1.5.4 Work Potential 10
1.5.5 Principle of Minimum Potential Energy 11
1.5.6 Hamilton's Principle 11
1.5.7 Rayleigh–Ritz Method 13
1.5.8 Rayleigh–Ritz Method with Hamilton's Principle 17
1.6 Galerkin's Method 19
1.7 Linearity, Symmetry, and Positive Definiteness 26
Computer Programs 27
Problems 27

2 Gaussian Elimination 35

2.1 Introduction 35
2.2 Basic Concepts in Matrix Algebra 35
2.2.1 Simultaneous Equations 35
2.2.2 Positive Definiteness 36
2.2.3 Positive Semi-Definiteness 36
2.2.4 Diagonally Dominant Matrix 37

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

2.2.5 Eigenvalues and Eigenvectors 37
2.2.6 Norms 38
2.3 Gaussian Elimination 39
2.3.1 General Algorithm for Gaussian Elimination 40
2.4 LU Factorization 42
2.4.1 Forward and Backward Substitution 43
2.5 Pivoting 44
2.6 Symmetric, Banded Matrices 46
2.7 Cuthill–McKee Algorithm to Reduce Bandwidth 50
2.8 Skyline (Variable-Band) Matrices 53
2.8.1 Storage of a Symmetric Skyline Matrix 54
2.8.2 Gaussian Elimination with Column Reduction 54
2.8.3 Note on Pivoting with Ordering 58
2.9 General Sparse Symmetric Matrices 58
2.9.1 Pivoting Based on the Minimum Degree Algorithm 60
2.10 Conjugate Gradient Method for Equation Solving 62
2.10.1 Conjugate Gradient Algorithm 62
2.11 Frontal Method for FE Matrices 63
2.11.1 Connectivity and the Prefront Routine 64
2.11.2 Element Assembly and Consideration of Specified dof 64
2.11.3 Elimination of Completed dof 65
2.11.4 Backward Substitution 65
2.11.5 Consideration of Multipoint Constraints 66
2.12 Using In-Built MATLAB® Routines 66
Computer Programs 67
Problems 67

3 One-Dimensional Elasticity 71

3.1 Introduction 71
3.2 Equations of Elasticity in 1D 71
3.3 Finite Element Modeling 74
3.3.1 Element Division 74
3.3.2 Numbering Scheme 74
3.4 Local Coordinates, Shape Functions, [N] and [B] Matrices 75
3.4.1 ξ - or Intrinsic Coordinates 76
3.4.2 Shape or Interpolating Functions N_1 and N_2 77
3.4.3 Displacement Interpolation 77
3.4.4 Admissibility Conditions 79
3.4.5 The Strain–Displacement Matrix [B] 79
3.5 Element Stiffness and Load Matrices from Potential Energy 80

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

3.6 Assembly of the Global Stiffness Matrix and Load Vector 83
3.7 SPC-Type Boundary Conditions 85
3.7.1 Elimination Technique 85
3.7.2 Decoupling Technique 87
3.7.3 Penalty Function Technique 88
3.8 Boundary Conditions and Singularity of [K] 89
3.9 MPC-Type Boundary Conditions 98
3.9.1 Master–Slave Elimination Technique 98
3.9.2 Penalty Function Technique 100
3.10 Banded Assembly of Stiffness Matrix 101
3.11 Convergence Aspects and Adaptivity 102
3.11.1 Monotonic Convergence in Energy 102
3.11.2 Rate of Convergence 103
3.11.3 Adaptivity 105
Computer Programs 108
Problems 109

4 One-Dimensional Heat Conduction 118

4.1 Introduction 118
4.2 Governing Differential Equation 118
4.2.1 Boundary Conditions 120
4.3 Two-Node Linear Element 121
4.4 Galerkin's Approach 123
4.5 Handling SPC Boundary Conditions 125
4.5.1 BC $T_1 = T_0$ via Elimination Technique 126
4.5.2 BC $T_1 = T_0$ via Decoupling Technique 126
4.5.3 BC $T_1 = T_0$ via Penalty Function Technique 126
4.6 One-Dimensional Heat Transfer in Thin Fins 129
4.6.1 Derivation of Matrices via Galerkin's Approach 131
4.7 Convergence and Adaptivity 136
4.8 Transient Heat Conduction 137
4.8.1 Weak Form 138
4.8.2 Derivation of Matrices 138
4.9 A Discussion of Integration Techniques 143
4.9.1 Mesh Size 144
4.9.2 Explicit Methods 144
4.9.3 Implicit Methods 145
4.9.4 Mode Superposition 146
Computer Programs 147
Problems 148

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

5 Trusses 152

5.1 Introduction 152
5.2 Two-Dimensional Trusses 152
5.2.1 Local and Global Coordinate Systems 152
5.2.2 Formulas for Calculating ℓ and m 154
5.2.3 Element Stiffness Matrix 155
5.2.4 Stress Calculation 156
5.2.5 Temperature Load Vector 162
5.3 Three-Dimensional Trusses 164
5.4 Problem Modeling and Boundary Conditions 166
5.4.1 Incorrect BCs Causing Singularity 166
5.4.2 Symmetry 167
5.4.3 Anti-symmetry 167
5.4.4 Multipoint Constraints 168
5.5 Assembly of Global Stiffness Matrix for the Banded and Skyline Solutions 169
5.5.1 Assembly for Banded Solution 170
5.5.2 Assembly for Skyline Solution 171
Computer Programs 172
Problems 173

6 Two- and Three-Dimensional Elasticity 183

6.1 Introduction 183
6.2 Equations of Elasticity in 3D 183
6.2.1 Displacements and External Loads 183
6.2.2 Stresses and Equilibrium 184
6.2.3 Strain–Displacement Relations 187
6.2.4 Stress–Strain Relations 188
6.2.5 Compatibility Equations 189
6.2.6 Potential Energy and Work Potential 190
6.2.7 Weak Form Associated with Galerkin's Method 191
6.3 Equations of Elasticity in 2D 191
6.3.1 Plane Stress 191
6.3.2 Plane Strain 191
6.4 Numbering Scheme, Element Library, Shape Functions, and Notation Used 195
6.4.1 Three-Node Linear Triangular Element 197
6.4.2 Four-Node Quadrilateral Element 200
6.4.3 Degenerate Quad 201
6.4.4 Nine-Node Quadrilateral 201
6.4.5 Eight-Node Quadrilateral 204
6.4.6 Six-Node Triangle 205

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

6.4.7 Four-Node Tetrahedral 206
6.4.8 Eight-Node Hexahedral 207
6.4.9 Admissibility Conditions 208
6.5 The Strain–Displacement Matrix [B] 208
6.5.1 Two-Dimensional Elements 208
6.5.2 Three-Dimensional Elements 214
6.6 Element Stiffness Matrix and Load Vectors 214
6.7 Numerical Integration 217
6.7.1 One-Dimensional Integrals 217
6.7.2 Two- and Three-Dimensional Integrals for Quadrilateral and Hexahedral Elements 219
6.7.3 Stiffness Integration 220
6.7.4 Numerical Integration for Triangular and Tetrahedral Elements 221
6.7.5 Stress Calculations 222
6.8 Modeling and Boundary Conditions 226
6.8.1 Rectangular Plate 226
6.8.2 Octagonal Pipe (Inclined Roller) 226
6.8.3 Rigid Connection 227
6.8.4 Part with Pin Connection Subject to an Impact Load 228
6.8.5 Three-Dimensional Model of a Cylinder on Bearings 229
6.8.6 Pyramid-Shaped Part 229
6.8.7 Some General Comments on Meshing 230
6.9 Assembly for Sparse Solvers 231
6.9.1 Banded Assembly 231
6.9.2 Skyline Assembly 233
6.9.3 Sparse Assembly and Solution via MATLAB® Routines 233
6.9.4 Frontal Assembly 234
6.10 Convergence Aspects and Adaptivity 237
6.10.1 Monotonic Convergence in Energy 237
6.10.2 Adaptivity 241
6.10.3 Stress Singularities 241
Computer Programs 242
Problems 242

7 Axisymmetric Problems in Elasticity 259

7.1 Introduction 259
7.2 Equations of Elasticity in Axisymmetric Analysis 259
7.2.1 Displacements and External Loads 259
7.2.2 Stresses and Equilibrium 260
7.2.3 Strain–Displacement Relations 261

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

7.2.4 Stress–Strain Relations 262
7.2.5 Potential Energy and Work Potential 263
7.3 Derivation of Strain–Displacement Matrix [B_a] 263
7.4 Element Stiffness Matrix and Load Vectors 266
7.4.1 Formulas for Constant Body Force and Constant Traction on a Three-Node Triangular Element 267
7.5 Problem Modeling and Boundary Conditions 271
7.5.1 Arresting Rigid-Body Motion 271
7.5.2 Cylinder Subjected to Internal Pressure 272
7.5.3 Infinite Cylinder 272
7.5.4 Press-Fit on a Rigid Shaft 273
7.5.5 Press-Fit on an Elastic Shaft 274
7.5.6 Belleville Spring 274
7.5.7 Thermal Stress Problem 275
7.5.8 Notation Used in Commercial Programs 275
Computer Programs 283
Problems 284

8 Two- and Three-Dimensional Heat Conduction and Other Scalar Field Problems 293

8.1 Introduction 293
8.2 Steady-State Heat Transfer 293
8.2.1 Differential Equation 296
8.2.2 Boundary Conditions 296
8.2.3 Finite Element Interpolation for 2D Elements 297
8.2.4 Three-Node Triangular Element 297
8.2.5 Four-Node Quadrilateral Element 298
8.2.6 The [B] Matrix 298
8.2.7 Element Matrices Derived from Galerkin's Approach 300
8.2.8 Specific Expressions for the Three-Node Triangular Element 303
8.2.9 Preprocessing for Program *Heat2D* 308
8.2.10 Two-Dimensional Fins 308
8.3 Transient Heat Conduction, Including Axisymmetric Problems 310
8.4 Torsion, Potential Flow, Seepage, Electric and Magnetic Fields, Fluid Flow in Ducts, and Acoustics 314
8.4.1 Torsion Problem 314
8.4.2 Potential Flow 317
8.4.3 Seepage 318
8.4.4 Fluid Flow in Ducts 319
8.4.5 Electrical and Magnetic Field Problems 321
8.4.6 Acoustics 325
Computer Programs 329
Problems 329

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

9 Elements in Bending 340

9.1 Introduction 340
9.2 Beams 340
9.2.1 Potential Energy 342
9.2.2 Finite Element Formulation 343
9.2.3 Hermite Shape Functions 343
9.2.4 Element Stiffness Matrix 346
9.2.5 Load Vector 346
9.2.6 Shear Force and Bending Moment 349
9.3 Beams on Elastic Supports 354
9.4 Plane Frames 355
9.5 Three-Dimensional Frames 361
9.6 Modeling Aspects 365
9.6.1 Symmetry 365
9.6.2 Pin Joint 366
9.7 Thin Plates in Bending 366
9.7.1 Potential Energy 368
9.7.2 Nine-dof Triangular Element 368
9.7.3 Absence of a C^1 Continuous Displacement Field for a Nine-dof Element 369
9.7.4 The DKT Element 370
9.7.5 Converting Distributed Loads to Equivalent Point Loads in the DKT Element 371
9.8 Folded Plates 374
9.9 Thermal Stresses 375
Computer Programs 376
Problems 376

10 Structural Vibration 384

10.1 Introduction 384
10.2 Formulation 384
10.2.1 Hamilton's Principle 384
10.2.2 Solid Body with Distributed Mass 386
10.2.3 Natural Frequencies and Mode Shapes: the Eigenvalue Problem 387
10.3 Consistent Element Mass Matrices 389
10.3.1 One-Dimensional or Bar Element 389
10.3.2 Truss Element 389
10.3.3 Three-Node Triangular or CST Element 391
10.3.4 Isoparametric Elements in 2D and 3D Elasticity 392
10.3.5 Tetrahedral Element 393
10.3.6 Hexahedral Element 393
10.3.7 Axisymmetric Triangular Element 394

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

10.3.8 Beam Element 395
10.3.9 Frame Element 395
10.3.10 DKT Plate Bending Element 396
10.4 Lumped Element Mass Matrices 397
10.4.1 Technique 1 397
10.4.2 Technique 2 399
10.5 Forced Response Problems 408
10.5.1 Mode Superposition 408
10.5.2 Direct Integration Techniques 410
10.6 Guyan Reduction 413
10.7 Methods for Solving the Eigenvalue Problem 415
10.7.1 Properties of Eigenvectors 416
10.7.2 Characteristic Polynomial Method 416
10.7.3 Vector Iteration Methods 417
10.7.4 Shifting 419
10.7.5 Orthogonal Space 419
10.7.6 Transformation Methods 420
10.7.7 Generalized Jacobi Method 420
10.7.8 Tridiagonalization and the Implicit Shift Approach 425
10.7.9 Bringing the Generalized Problem to the Standard Form 425
10.7.10 Tridiagonalization 426
10.7.11 Implicit Symmetric QR Step with Wilkinson Shift for Diagonalization 429
10.8 Buckling Analysis 430
Computer Programs 434
Problems 434

11 Miscellaneous Topics 444

11.1 Introduction 444
11.2 Linear Orthotropic Materials 444
11.2.1 Temperature Effects 447
11.3 Nonlinear Heat Conduction Problems in 1D 450
11.3.1 Temperature-Dependent Thermal Conductivity 450
11.3.2 Load Increment Loop 452
11.3.3 Nonlinear Elastic Material Under Small Strain 458
11.3.4 Physical Interpretation of the Newton–Raphson Method 459
11.4 Additional Nonlinear Problems 460
11.4.1 Elastoplastic Problem in 1D 461
11.4.2 Buckling Load Determination via Nonlinear Analysis 462
11.4.3 General Comments 464
Problems 465

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

12 Preprocessing and Postprocessing 468

12.1 Introduction 468
12.2 Mesh Generation for 2D Plane Problems 468
12.2.1 Region and Block Representation 468
12.2.2 Block Corner Nodes, Sides, and Subdivisions 469
12.2.3 Generation of Node Numbers 470
12.2.4 Generation of Coordinates and Connectivity 472
12.2.5 Examples of Mesh Generation 473
12.2.6 Mesh Plotting 474
12.2.7 Data Handling and Editing 474
12.3 Filling a 3D Region with Tetrahedral Elements 475
12.4 Postprocessing 479
12.4.1 Deformed Configuration and Mode Shape 479
12.4.2 Contour Plotting 479
12.4.3 Nodal Values from Known Constant Element Values for a Triangle 481
12.4.4 Least-Squares Fit for a Four-Node Quadrilateral 482
12.5 Conclusion 484
Computer Programs 486
Problems 486

Appendix List of Symbols and Computer Code Variables 489
Bibliography 490
Index 493

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

Preface

Overview

Finite elements ("FE" or "FEA") is a numerical tool used for analyzing problems involving stress analysis, heat and fluid flow, resonance frequencies, mode shapes, and more. Irregularly shaped domains and various materials can be incorporated. This book deals with a variety of topics in a manner that integrates theory, algorithms, modeling, and computer implementation. Many solved examples reinforce this pedagogy, along with end-of-chapter problems, in-house source codes for multiple platforms, and a solutions manual for the instructor. Topics include energy and Galerkin approaches, equation solving with sparsity, elasticity, heat conduction and other scalar field problems, vibration, and pre- and postprocessing. The variety of topics dealt with enables the book to be used as a text in various engineering disciplines at the senior undergraduate or first-year graduate level. The book can also serve as a learning resource for practicing engineers.

Computer implementation, which parallels the theory, has helped to explain the material in the text with greater insight and accuracy. The programs can be tried out by students, researchers, and practicing engineers. In this fifth edition, the pedagogy is improved by: (1) increasing the number of solved examples from 49 in the earlier edition to 85 in the current one; (2) the variety and details in each solved example, including writing of specific code (contained in the website noted below); (3) new topics, including Galerkin and energy approaches, singularity of stiffness, convergence, adaptivity, handling time-dependent problems, sparsity in Gaussian elimination and in finite element assembly, using MATLAB® routines, and plate bending; and (4) adding in-house Python and MATLAB® programs.

Highlights Pertaining to Computer Programs Provided

- In-house programs have been developed in MATLAB®, Python, Fortran, C, JavaScript, and Excel VBA. These use input–output files with banded, skyline, and frontal solvers. The computer programs are provided on the website: www.cambridge.org/chandrupatla-belegundu5e and are organized in subdirectories as: \Matlab, \Python, \Fortran, \C, \JavaScript, \ExcelVBA, \EXAMPLES, \EXTRAS.
- The \EXAMPLES subdirectory contains input (.inp) files that interface to the computer programs. \ExcelVBA and \Javascript programs have integrated data.

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

- The \EXTRAS contain MATLAB® files used in the solved examples. These provide understanding, and also can be readily rewritten in other languages as they follow the theory in a transparent manner, and use in-built routines sparingly.
- A Readme.txt file is provided in the main directory. At the end of each chapter the names of programs pertinent to that chapter are listed.
- A solutions manual is provided to instructors.
- Appendix 1 contains a list of symbols and computer code variables.

Highlights Pertaining to Book Chapters

Chapter 1: Energy and Galerkin Approaches

This first chapter focuses on two main approaches: energy and Galerkin. Energy is used on elasticity problems, while Galerkin is used in heat conduction and other scalar field problems. Ritz basis functions and admissibility aspects are discussed. More detailed explanations of essential and natural boundary conditions and applications to time-dependent problems are presented. A section is given on properties of a differential operator that ties in with the solved examples. 2D/3D elasticity equations are deferred to Chapter 6. End-of-chapter problems focus on the two keystone approaches.

Chapter 2: Gaussian Elimination

Gaussian elimination is discussed with LU decomposition in general, banded, skyline solvers, along with algorithms and computer programs. Pivoting has also been added. The Cuthill–McKee algorithm for reducing the bandwidth and profile of a matrix is presented (code given in the MATLAB® subdirectory). Use of in-built MATLAB® routines covering various aspects including LU decomposition, bandwidth, and profile reduction and sparse solution are explained.

Chapter 3: One-Dimensional Elasticity

The advantage of piecewise linear basis functions over global polynomials used in Rayleigh–Ritz methods is emphasized. Handling SPC-type (single point) boundary conditions by elimination, penalty, and decoupling techniques is presented. Handling MPC-type (multi-point) boundary conditions by master–slave and penalty function techniques is presented. A section on singularity of stiffness is added which is of practical importance to the user. In a new section, mesh convergence and adaptivity with h-elements are rigorously presented.

Chapter 4: One-Dimensional Heat Conduction

This is a new chapter. Earlier, it was part of a chapter on 1D and 2D heat conduction plus other scalar field problems. Based on reviewer feedback, an early chapter on 1D heat conduction has made the first four chapters of the text a self-contained foundation for the

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

linear finite element method. Galerkin's method is used and explained in detail. Time-dependent problems are formulated, and solutions via implicit, explicit, and mode superposition techniques are presented.

Chapter 5: Trusses

Improper modeling of boundary conditions causing singularity of stiffness is explained along with modeling symmetry and anti-symmetry. Multipoint constraint modeling and solution using a technique given in Chapter 3 is presented.

Chapter 6: Two- and Three-Dimensional Elasticity

In earlier editions, three-node triangular elements, four-node and higher-order elements, and 3D tetrahedral and hexahedral elements were covered in separate chapters. Owing to the considerable overlap, and since the element matrices can be derived using the same general expressions, namely $\mathbf{B} = \mathbf{AG}$, these have been unified in this chapter. Numerical integration fits naturally in this unified treatment. The chapter begins with the equations of elasticity in 3D and 2D with the Rayleigh–Ritz technique applied to a problem, which is later solved using the FE method. Modeling considerations and integration of the codes (i.e., assembly process) with both in-house and MATLAB® sparse solvers are discussed. Convergence aspects and adaptivity are presented rigorously, building on the material given in Chapter 3. Singularities specific to 2D/3D problems are outlined.

Chapter 7: Axisymmetric Problems in Elasticity

Earlier, the material was closely interconnected with use of three-node triangular elements. Here, the development is general in that any type of element (three- or four-node) can be used. Numerical integration fits more naturally and clearly in the revised treatment. Via a solved example, detailed treatment of a press-fit involving MPCs is given with convergence of results using mesh refinement.

Chapter 8: Two- and Three-Dimensional Heat Conduction and Other Scalar Field Problems

Changes here are reflected by the details involving solved examples, as for example involving a contour integral to evaluating the impedance of a coaxial element, and a time-dependent heat conduction problem in cylindrical coordinates.

Chapter 9: Elements in Bending

With regard to beams, additions pertain to plotting of beam deflections using Hermite shape functions and plotting of the bending moment diagram, as part of the postprocessing step.

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

In addition to beams and 2D/3D frames, bending of thin plates is presented. Details pertaining to the DKT (discrete Kirchoff triangle) element are given. Proper handling of distributed loads using the element is discussed, with solved examples. This element is used for solving a thermal stress problem, and later used for vibration analysis in Chapter 10.

Chapter 10: Structural Vibration

Lumped mass matrices are discussed in more detail and solved examples compare results using consistent and lumped masses. Mode superposition is introduced for handling forced vibration problems. Other topics relating to Guyan reduction and methods for solving the eigenvalue problem are retained. A new section on eigen buckling analysis with truss elements is added.

Chapter 11: Miscellaneous Topics

This is a new chapter. Orthotropic materials are discussed – this was part of the earlier edition. Introduction to nonlinear FEA analysis is new. Basic concepts of the Newton–Raphson method and tangent stiffness applied to 1D heat conduction and to 1D nonlinear elastic material under small strain are presented. Explicit integration is shown in a nonlinear transient heat conduction problem via a solved example. A nonlinear analysis of a simple truss structure is given to estimate the buckling load, which is then compared to that obtained from solving the eigenvalue problem.

Chapter 12: Preprocessing and Postprocessing

This chapter deals with 2D mesh generation and contour plotting and is the same as in the earlier edition.

How to Use this Book

This book can be navigated by the instructor or reader based according to the following:

- Chapters 1–4 form the foundation of the linear FE method covering elasticity and heat conduction (scalar field).
- Chapter 11 covers orthotropic materials and offers an introduction to nonlinear problems.
- Chapters 5, 6, 7, and 9 focus on structural problems.
- Chapter 8 focuses on 2D and 3D scalar fields.
- Chapter 10 focuses on vibration.
- Chapter 12 may be covered together with Chapter 6.

We are grateful to the students at our respective institutions for motivating us to further develop this book. Professor Chandrupatla would like to thank Dr. Ratneshwar Jha,

© in this web service Cambridge University Press & Assessment www.cambridge.org

CAMBRIDGE

Cambridge University Press & Assessment
978-1-108-84141-2 — Introduction to Finite Elements in Engineering
5th Edition
Tirupathi Chandrupatla , Ashok Belegundu
Frontmatter
More Information

department head, for encouragement received. He also wishes to express his gratitude to Professor J. Tinsley Oden of the University of Texas, who provided the inspiration needed. Professor Belegundu would also like to thank Dr. Karen Thole, department head, for her strong encouragement in this effort, and Professor Subramaniam D. Rajan at Arizona State University for many discussions. We are thankful to the many reviewers of this book for their excellent feedback.

© in this web service Cambridge University Press & Assessment www.cambridge.org